

Cricket Performance Player Prediction

Predicting cricket player performance from previous match records using Ridge, Lasso and Elastic Net regression.

TEAM

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THE TASK

Problem Statement

REGRESSION TASK

Estimate how many runs a batter will score in an innings from their recent form and match involvement — framed as a supervised regression problem on continuous output.

TARGET VARIABLE

runs_x — runs scored in a single innings. Range 0–329, mean 23.8, standard deviation 28.6 across 46,263 innings.

WHY MANUAL ANALYSIS FAILS

Performance depends on balls faced, boundaries, batting order, format, opposition and recent form at once. Weighing these by eye is subjective and does not scale across 946 players.

WHY IT MATTERS

A reliable estimate supports team selection, fantasy scoring and workload planning — and shows which performance factors actually carry predictive signal.

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Models Compared

Ridge · Lasso ·

4

Evaluation Metrics

MAE · MSE · RMSE · R^2

Project Objectives

01

Analyse historical records

Explore 46,300 batting innings across T20, ODI and Test formats.

02

Identify key drivers

Measure which involvement and form variables correlate with runs.

03

Clean & preprocess

Handle missing values, verify types, scale numeric predictors.

04

Engineer form features

Create features based on a player's **previous performances**, such as career averages and recent-form averages.

05

Build regression models

Train Ridge, Lasso and Elastic Net on an identical pipeline.

06

Evaluate

Score every model on MAE, MSE, RMSE and R^2 .

Motivation

01

Form Changes

A career average built over 10 years cannot notice that a player has failed in his last five innings. Recent form and career form are different pieces of evidence.

02

One statistic is not enough

Runs, strike rate, boundaries, balls faced and batting position each describe a different part of a batter. Selection debates usually quote only one of them.

03

Records contain patterns

Across 46,300 innings the same signals repeat: batters who face more balls and hit more fours historically also score more in the next innings.

04

ML combines predictors

Regression can weigh all sixteen historical signals together, instead of a human eyeballing one column at a time.

“Instead of simply asking whether a player is good, the project attempts to quantitatively estimate player performance using evidence from previous records.”

Dataset Description

RAW INNINGS RECORDS

46,300

PLAYERS MODELLED

946

SEASONS COVERED

2004–2021

SOURCE FILES

bat.csv · ball.csv · match.csv

bat.csv is the modelling table: one row per batter per innings, joined to match-level context.

ball.csv holds bowling figures and match.csv holds innings totals, run rates and team identifiers.

- runs_x — runs scored (target)
- balls — deliveries faced
- fours / sixes — boundaries hit
- order — position in batting line-up
- strike_rate — scoring efficiency
- matchtype — T20 / ODI / Test
- over, run_rate — innings context
- start_date, player_id — form tracking keys

TARGET VARIABLE

runs_x — innings runs

TARGET VARIABLE — runs_x

Innings-level batting performance

0–329 runs per innings

Mean: 23.8 runs

Median: 13 runs

Used as the continuous target for regression

Dataset Snapshot

NUMERICAL FEATURES

- runs_x
- balls
- strike_rate
- fours
- sixes
- order
- over
- run_rate
- fall_of_wicket_over

CATEGORICAL FEATURES

- name_x — player
- playing_role
- matchtype
- how_out
- title — innings
- team_id
- opp_team_id

DATE / KEY FEATURES

- start_date
- player_id
- match_id
- scorecard_id
- series_id
- match_number

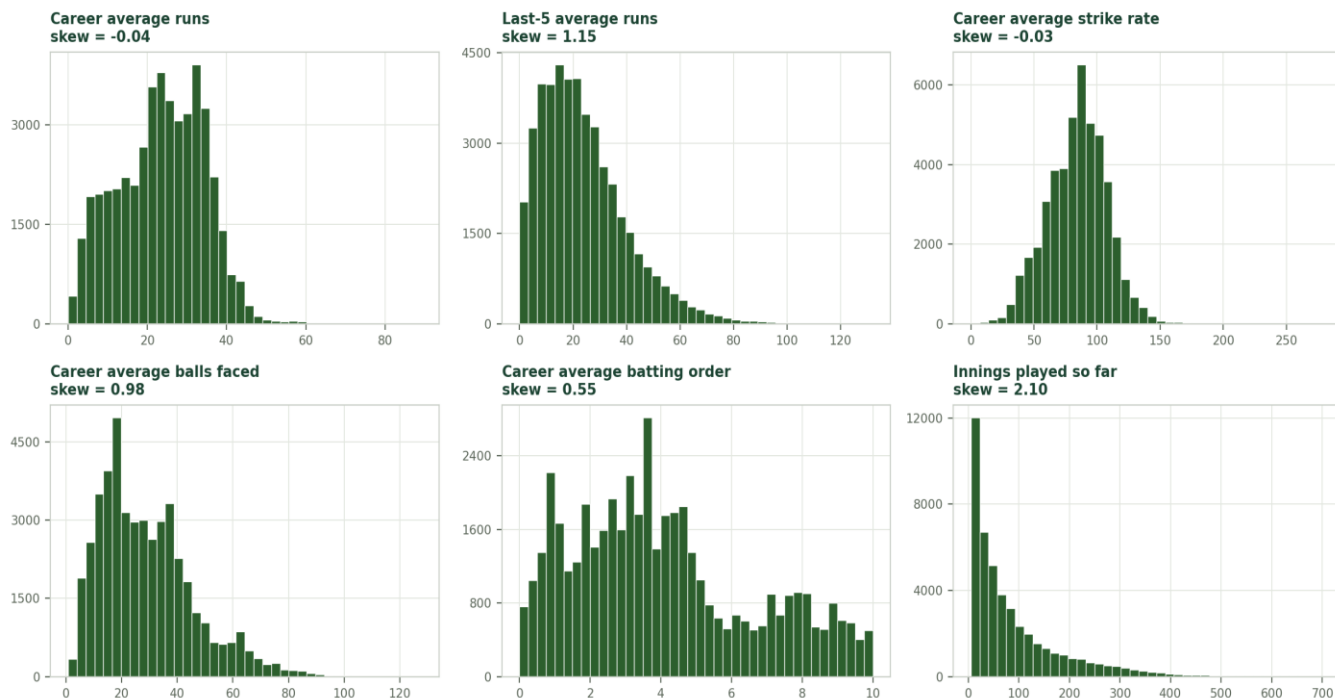
This slide shows the different types of variables in our dataset. We have numerical features, categorical features, and date or identifier columns. Categorical variables such as match type are encoded so that the machine learning models can process them. Identifiers are mainly used for tracking and grouping rather than prediction

Data Cleaning & Preprocessing

Problem	Observation in our data	Handling
Missing values	playing_role 1,543, match_number 635, matchtype 37, name_y 13, how_out 9. ball.csv: fours/sixes/zeros 36,134 (99.96%). match.csv: match_number 75, name 2, matchtype 4	Drop near-empty columns; label missing matchtype as “unknown” rather than guessing
Duplicate records	0 fully duplicated rows in all three files. 0 duplicates on (player_id, scorecard_id) — each player appears once per innings	Nothing to remove; uniqueness confirms one row = one innings
Data types	start_date read as text; match_number stored as float; index column imported as “Unnamed: 0”	Convert start_date to datetime (essential for chronological ordering); drop the stray index column
Identifier columns	player_id, match_id, scorecard_id, series_id, team_id, opp_team_id, match_detail_id are arbitrary numbers	Used only for grouping and joining — never as numeric predictors
Invalid values	828 innings with balls = 0 (all have 0 runs and SR 0 — valid “out without facing”). strike_rate reproduces runs/balls×100 exactly in every checkable row; boundary runs never exceed total runs. how_out contains HTML fragments (†, &)	No impossible statistics found. Keep 0-ball innings; decode HTML entities in how_out

EDA — Feature Distributions

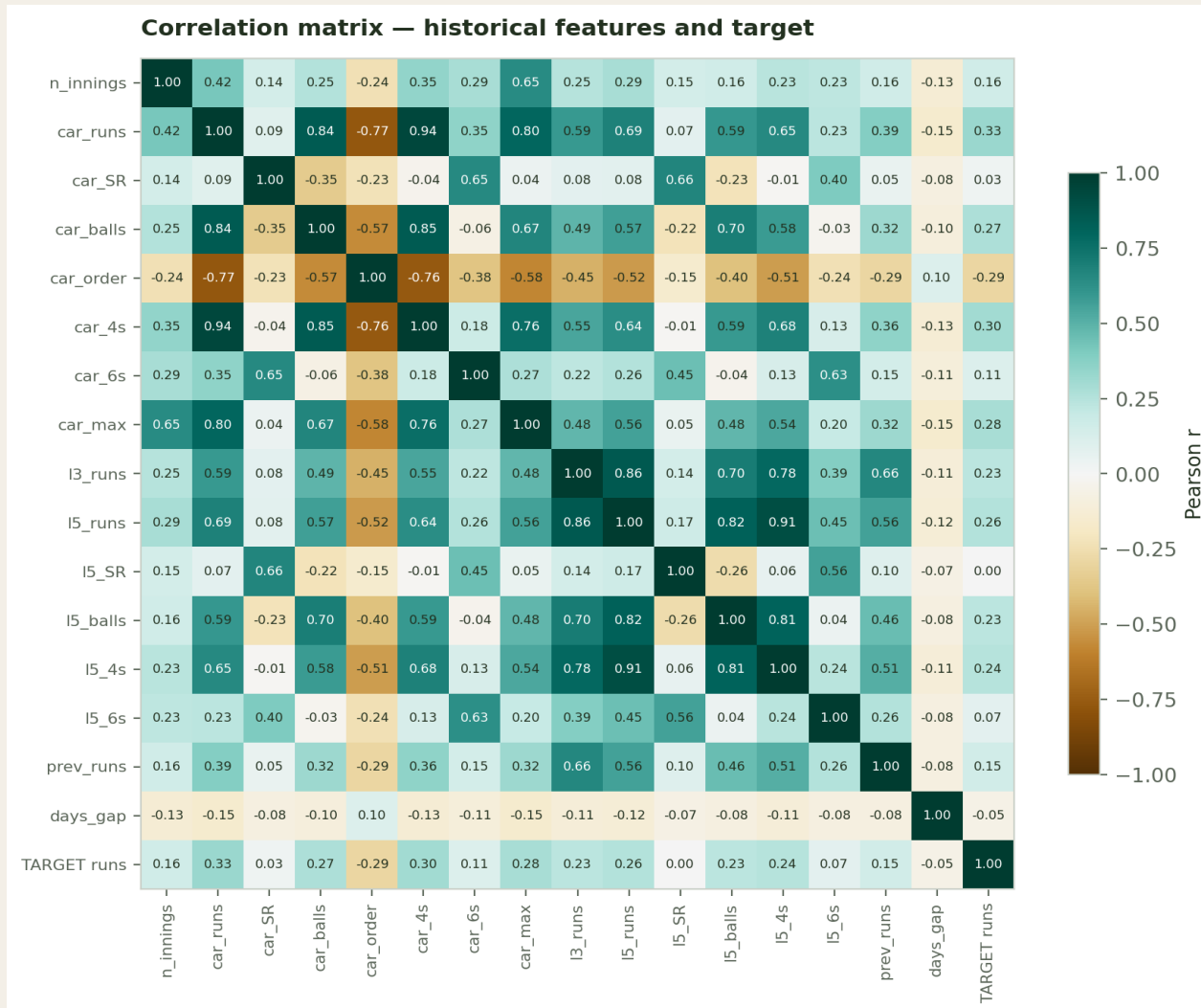
Distributions of the main historical (pre-match) features



Graph	What is it showing?	X-axis	Y-axis	Skew
Career Average Runs	Distribution of players' average runs scored across their previous career innings	Career average runs	Number of observations	-0.04
Last-5 Average Runs	Distribution of players' average runs in their most recent 5 innings, showing recent form	Average runs in last 5 innings	Number of observations	+1.15
Career Average Strike Rate	Distribution of players' average strike rate across previous innings, showing scoring efficiency	Career average strike rate	Number of observations	-0.03
Career Average Balls Faced	Distribution of players' average number of balls faced per innings, showing batting involvement	Career average balls faced	Number of observations	+0.98
Career Average Batting Order	Distribution of players' average batting position, showing they generally bat	Career average batting position	Number of observations	+0.55
Innings Played So Far	Distribution of how many previous innings a player has played before the current prediction	Previous innings count	Number of observations	+2.10

Why this matters before modelling: Ridge and Lasso penalise the size of the coefficients. A feature measured in the hundreds needs a small coefficient and a feature measured in units needs a large one, so an unscaled penalty punishes the small-scale features unfairly. → StandardScaler first, then Ridge / Lasso.

Correlation Analysis & Multicollinearity



Strongly correlated predictor pairs

Pair	r
career_avg_runs ↔ career_avg_fours	+0.94
last5_avg_runs ↔ last5_avg_fours	+0.91
last3_avg_runs ↔ last5_avg_runs	+0.86
career_avg_balls ↔ career_avg_fours	+0.85
career_avg_runs ↔ career_avg_balls	+0.84
career_avg_runs ↔ career_max_runs	+0.80
career_avg_runs ↔ career_avg_order	-0.77

Why this decides the algorithm

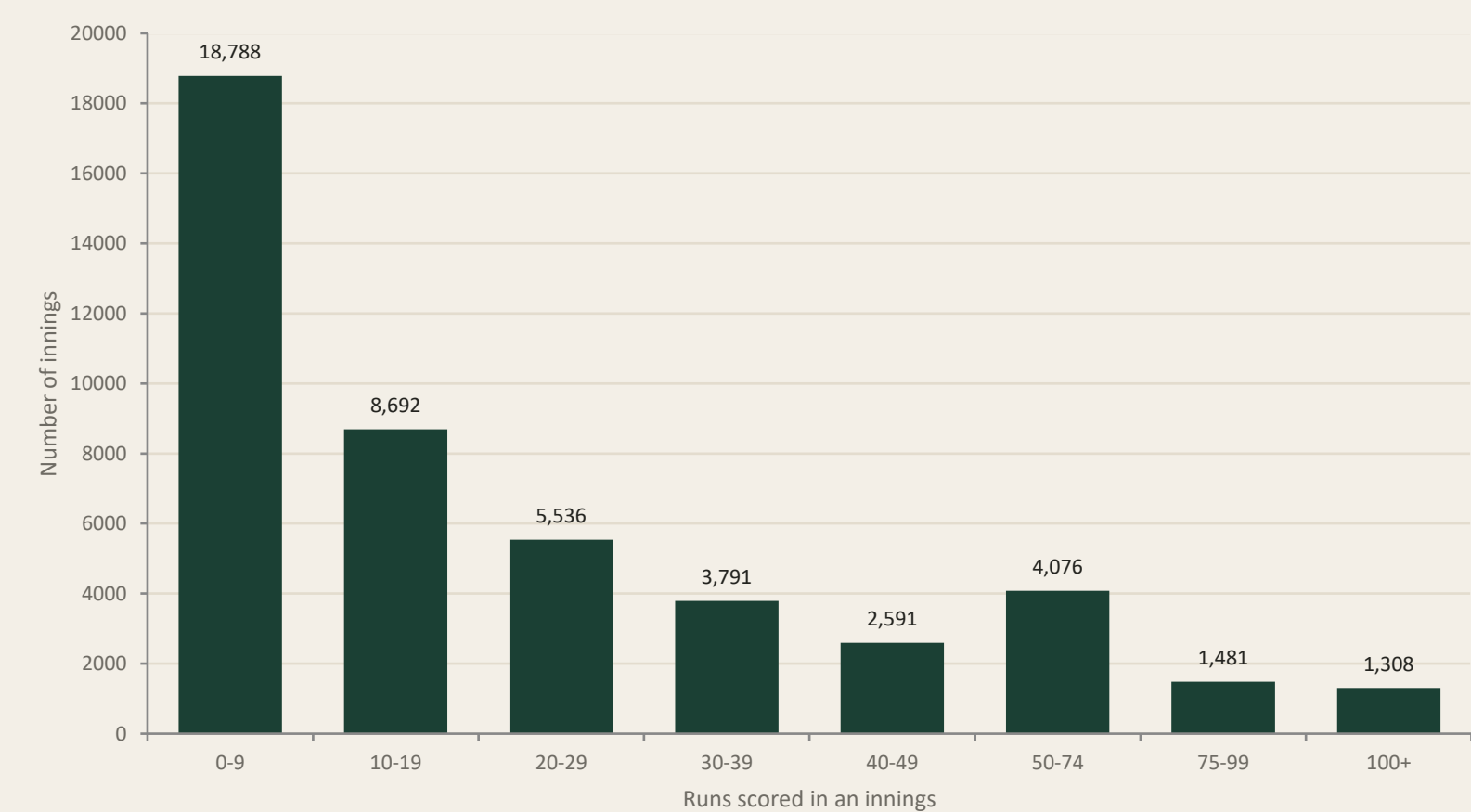
Correlated predictors (r up to 0.94)

Ordinary least squares cannot tell them apart

Coefficients become large, unstable, sign-flipping

Add a penalty on coefficient size

EDA — Distribution of Runs



runs — half of all innings finish at 13 runs or fewer.

Part	Meaning
X-axis	Runs scored in an innings
Y-axis	Number of innings/observations
Median = 13	Half of the innings have 13 runs or fewer
Single-digit scores	18,788 innings
Centuries (100+)	1,308 innings

Ridge Regression

L2 REGULARIZATION

Penalises the sum of squared coefficients

The loss becomes $RSS + \alpha \cdot \sum \beta^2$. Raising α shrinks every weight toward zero without ever reaching it, trading a little bias for a large reduction in variance.

$$RSS + \alpha \cdot \sum |\beta|^2$$

alpha = 1.0

Ridge Regression is Linear Regression with L2 regularization. Since our features have multicollinearity, Ridge adds a penalty based on the squared coefficients. This shrinks large coefficients, improves model stability, and helps control overfitting. In our implementation, alpha is set to 1.0, and the numerical predictors are scaled before training.

Handles multicollinearity

Balls, fours and sixes are strongly correlated; Ridge distributes weight across them instead of picking arbitrarily.

Stabilises coefficients

Small changes in the training sample no longer swing the fitted weights.

Keeps every feature

Nothing is dropped, so weak-but-real signals such as run rate still contribute.

Controls overfitting

Model complexity is limited without reducing the feature count.

Lasso Regression

LASSO — L1 REGULARIZATION

$$RSS + \alpha \cdot \sum |\beta|$$

The absolute-value penalty can force coefficients to exactly zero, so Lasso selects features as it fits. Configured with $\alpha = 0.01$.

ADVANTAGES & LIMITS

- Removes uninformative predictors outright
- Produces a sparser, easier-to-explain model
- Identifies which performance factors truly matter
- Can behave unstably when features are highly correlated

ADVANTAGES & LIMITS

- Selects features like Lasso, groups them like Ridge
- Keeps correlated predictors together rather than dropping one
- Two hyperparameters allow finer control of the trade-off
- Best all-round choice when predictors overlap in meaning

LARGEST STANDARDISED COEFFICIENTS (RIDGE)

+13.81

Fours

+12.49

Balls faced

+8.06

Sixes

+1.17

Career avg

-1.00

Overs

Model Evaluation Metrics

MAE

Mean Absolute Error

Average absolute gap between predicted and actual runs, in runs. Robust to the century tail.

LOWER IS BETTER

MSE

Mean Squared Error

Averages squared errors, so a single badly missed big innings costs far more than several small misses.

LOWER IS BETTER

RMSE

Root Mean Squared Error

Square root of MSE, returning the error to the original unit — runs — so it can be read directly.

LOWER IS BETTER

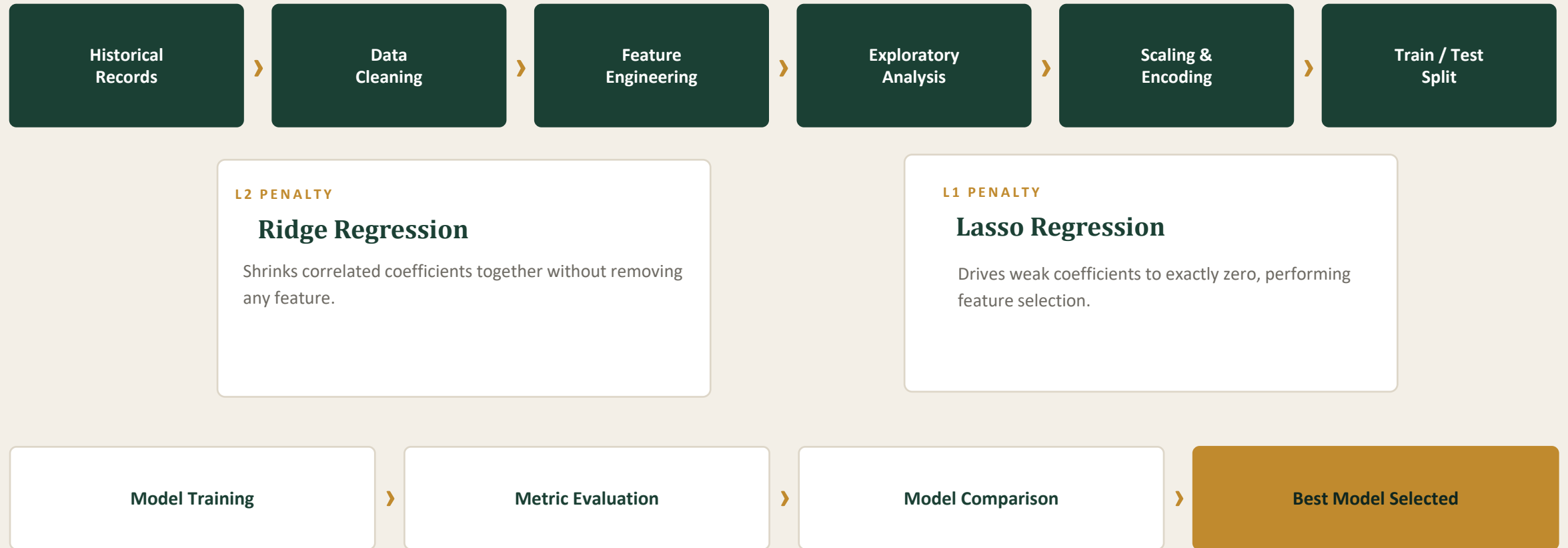
R²

R SQUARED

Share of the variance in innings runs the model explains. 1.0 is a perfect fit, 0 is no better than the mean.

HIGHER IS BETTER

Proposed Methodology



Conclusion

WHAT WAS BUILT

A working regression system for player performance

Historical match records were cleaned, enriched with leakage-free form features, and used to train Ridge, Lasso and Elastic Net models that estimate innings runs on unseen data with an average error of roughly four and a half runs.

WHAT WAS LEARNED

Regularization is a safeguard, not a boost here

On a wide, clean dataset with far more rows than predictors, L1 and L2 penalties barely alter the fit. Their value would appear with more features, smaller samples, or heavier collinearity.

FACTORS FOUND TO DRIVE PERFORMANCE

Balls faced

Opportunity

Fours & sixes

Scoring power

Batting order

Role proxy

Career average

Baseline skill

Recent form

Short-term trend

Match format

Scoring context

Thank You